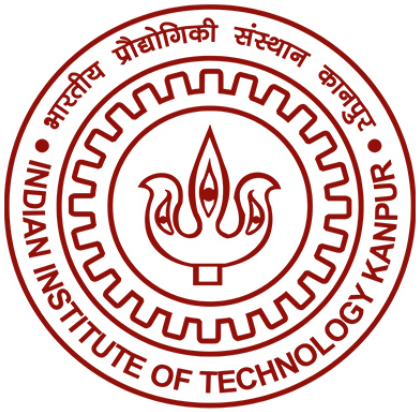


SPA617: RADIO ASTRONOMY LAB

RF transmission, reflection, standing waves

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Submission Date : 12-Aug-26

1 AIM

To understand signal propagation, reflection and standing wave formation in a coaxial transmission line at radio frequencies and to determine important cable parameters experimentally.

2 Objectives and Learning Outcomes

1. To determine the phase velocity and velocity factor of a signal in a coaxial cable, by measuring the frequency spacing between the consecutive minima produced the reflection caused in an open ended cable.

2. To study signal reflection from an open-ended cable and from a cable terminated with a $50\,\Omega$ load.

3. To estimate the magnitude of the reflection coefficient and verify that the coaxial cable is approximately matched with to a $50\,\Omega$ terminnation, indicating that its characteristic impedance is close to $50\,\Omega$.

4. To calculate the inductance and capacitor per unit length of the coaxial cable using the measured phase velocity and the approximate characteristic impedance of $50\,\Omega$.

5. To measure the insertion loss and attenuation per unit length of the coaxial cable.

3 Theory

A coaxial cable is a transmission line used to guide electromagnetic signals from one point to another. At radio frequencies, the signal propagates mainly as a TEM wave, and its velocity depends on the dielectric material between the conductors.

When a travelling wave reaches a load whose impedance is different from the characteristic impedance of the cable, part of the signal is reflected. The incident and reflected waves interfere with each other, producing maxima and minima in the measured signal amplitude. By varying the frequency, the phase of the reflected wave changes, causing a periodic variation in amplitude.

For an open-ended cable, the frequency difference between two consecutive maxima or minima can be used to determine the phase velocity:

$$v_p = 2l\Delta f$$

where l is the cable length and Δf is the frequency spacing.

The velocity factor is given by

$$VF = \frac{v_p}{c}.$$

The voltage reflection coefficient is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0},$$

where Z_L is the load impedance and Z_0 is the characteristic impedance of the cable. For an open circuit, $|\Gamma| = 1$, while for a matched load $Z_L = Z_0$, $\Gamma = 0$.

The inductance and capacitance per unit length of a low-loss transmission line are related to the characteristic impedance and phase velocity by

$$L' = \frac{Z_0}{v_p}, \qquad C' = \frac{1}{Z_0 v_p}.$$

The cable also introduces attenuation as the signal propagates. The insertion loss is determined by comparing the measured power with the cable to the reference power without the cable. The attenuation per unit length is then obtained by dividing the insertion loss by the cable length.

4 Equipments

- RF signal generator
- Digital storage oscilloscope(DSO)
- Spectrum analyser
- RF Tee connector
- Coaxial cable
- Short reference coaxial cable
- 50 Ω termination

5 Experimental setup

The RF signal generator is connected to a Tee connector. One arm of the Tee connector is connected to the coaxial cable under test, while the other arm is connected to the DSO. The end of the coaxial cable is terminated as required. The frequency of the signal generator is varied, and the signal amplitude on the DSO is observed. The frequencies corresponding to two consecutive minima are noted as f_1 and f_2.The frequency difference is calculated as

$$\Delta f = f_2 - f_1.$$

6 Result

Before starting the setup we carefully measured the coaxial cable which turned out to be 92.5 cm. Then after connecting all the cables with the Tee connector we observed the consecutive minimas using the Digital storage Oscilloscope (DSO). The frequency difference between those minimas was $\Delta f = f_2 - f_1 = (174 - 58)$ MHz.

After finding the frequency difference we can use that to find phase velocity.

$$v_p = 2\mathrm{l}\Delta f$$

$$v_p = 2 \times 92.5 \times 116 \text{ m/s}$$

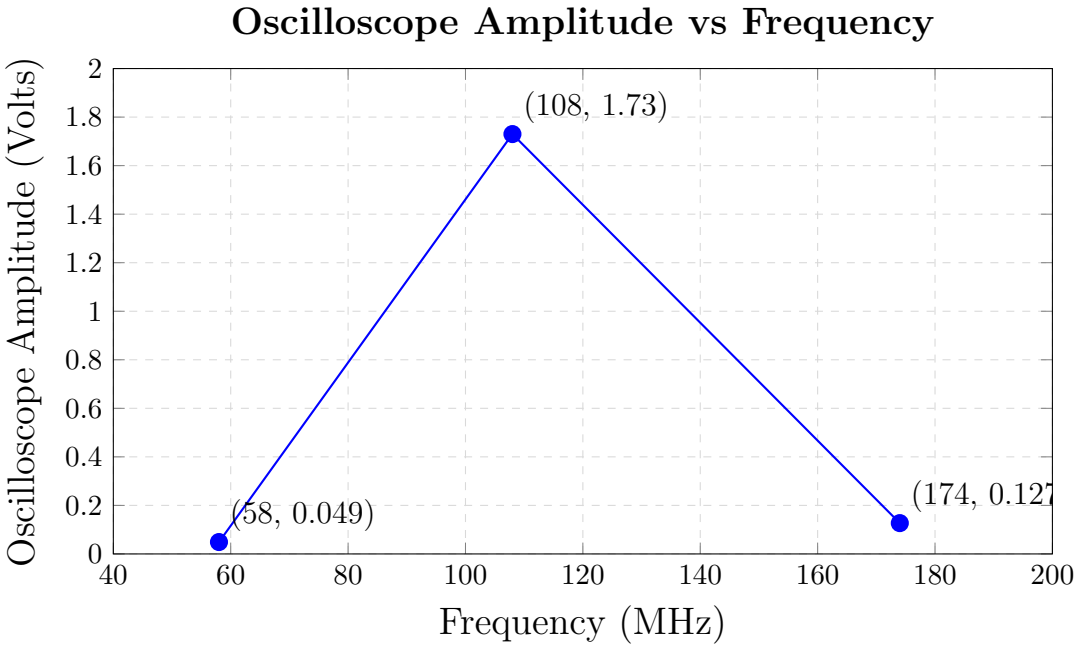


Figure 1: Oscilloscope amplitude as a function of frequency.

$$v_p = 2.146 \times 10^8 \text{ m/s}$$

Now with calculated phase velocity we can calculate the velocity factor-

$$\text{VF} = \frac{\text{v}_\text{p}}{\text{c}}$$

$$\text{VF} = \frac{2.146 \times 10^8}{3 \times 10^8}$$

$$\text{VF} = 0.715$$

In the next part of the experiment we checked the characteristic impedance. For that we first setup the Tee connector with a open circuit than re connected it with a 50 Ω termination at the far end. Retaining the same frequency as before to carefully determined the $V_{max,open}$ and $V_{min,open}$ for calculating the reflection coefficient. The same thing we have done with the 50 Ω termination load to note the difference in reflection coefficient.

Table 1: Reflection coefficient for open and 50 Ω load

Termination	$\mathbf{V}_{\text{max}}$ (V)	$\mathbf{V}_{\text{min}}$ (V)	$\|\mathbf{\Gamma}\|$
Open circuit	1.61	0.049	94.5%
50 Ω load	1.12	0.92	9.5%

With this substantial amount of reduction in reflection coefficient we can conclude that the cable is approximately matched with to the 50 Ω termination load. The calculated characteristic impedance turned out to be

$$Z_0 = \frac{1 - \Gamma}{1 + \Gamma}$$

$$Z_0 = 41.074 \ \Omega$$

Now using the measured value of v_p and Z_0 we can easily calculate the value of Inductance and Capacitance per unit length

$$\mathrm{L}' = \frac{\mathrm{Z_0}}{\mathrm{v_p}}$$

$$\mathrm{L}' = 23.33 \times 10^{-8}\ \mathrm{H/m}$$

and Capacitance per unit length will be

$$\mathrm{C}' = \frac{1}{\mathrm{Z_0} \times \mathrm{v_p}}$$

$$\mathrm{C}' = 0.00932 \times 10^{-8}\ \mathrm{F/m}$$

For the cable attenuation per unit length the spectrum analyser has been used to record reference power P_{ref} and P_{cable}. The calculated attenuation per unit length $\alpha = \frac{P_{ref} - Pcable}{l}$ dB/m is shown below

Table 2: Attenuation of the cable

Frequency(MHz)	$P_{ref}(dBm)$	$P_{cable}(dBm)$	$\alpha\ (dB/m)$
60	0.00	-0.65	0.703

Table 3: Experimental results for coaxial cable

Quantity	**Symbol**	**Measured/Calculated Value**
Cable length (m)	ℓ	92.5 cm
Frequency spacing (Hz)	Δf	116 MHz
Phase velocity (m/s)	v_p	2.146×10^8 m/s
Velocity factor	$\frac{v_p}{c}$	0.715
Approximate characteristic impedance (Ω)	Z_0	41.074 Ω
Inductance per unit length (H/m)	L'	233 nH/m
Capacitance per unit length (F/m)	C'	93.2 pF/m
Attenuation (dB/m)	α	0.703 dB/m

7 Conclusion

- In this experiment the characteristics of a coaxial cable were studied using the RF signal generator, Tee connector and DSO. The coaxial cable behaves as transmission line at high frequency and the signal doesn't travel simply from source to load. When load is not matched to the characterictic impedance of the cable, a part of the signal is reflected back towards the source. By varying the frequency and observing the consecutive minimas on the DSO the propagation velocity and other parameters of coaxial cable like velocity can be determined experimentally.

- We observed that changing the termination changes the reflected wave. In the case for short circuit the reflection coefficient is -1, which indicates the reflected wave is of same magnitude but with opposite phase. That is why it creates a standing wave producing successive minima and maxima. Same goes for the open ended circuit with only difference being that its maxima and minima position are interchanged with position of maxima minima for short circuit. In case of 50 Ω load the reflection coefficient becomes 0 hence no reflected wave for interference.

- As the velocity factor term stays constant for every cable, it can used to determine the length of a unknown cable by simply observing the frequency difference. In our case the given unknown cable's length turned out to be 305.7 cm which was well under the actual measured length (306.8 cm).

- The effect of termination load is also studied for calculating impedence and conductance per unit length.

- We observed a amplitude variation while connected to 50 Ω termination which shouldn't be the case. This indicates that our practical setup is not ideal perfectly matched transmission line. Dominant causes could be small impedance mismatch at the load or Tee connector also the coaxial cable has attenuation so the signal amplitude changes as it propagates.